SABANCI UNIVERSITY

CS412: MACHINE LEARNING

Homework 1

k-Nearest Neighbors

Overview

In this assignment, you will investigate the k-Nearest Neighbors (k-NN) classifier using the CIFAR-10 dataset. The assignment covers data splitting, exploratory analysis, preprocessing, hyperparameter selection, model evaluation, and error analysis.

Complete the provided **Jupyter Notebook template**, including all required code, results, visualizations, and written explanations.

A separate report is not required. All discussions, analyses, and explanations related to the homework must be included in the Jupyter Notebook. Please use the designated cells provided in the notebook for your written responses.

Submission Guidelines

- **Notebook:** Submit one completed `.ipynb` file. Include executed code cells, outputs, figures, and written answers. A separate PDF report is not required.
- **Filename:** `CS412-HW1-StudentID.ipynb`
- **Execution:** Run the notebook from beginning to end before submitting it. Make sure all required outputs are saved. The notebook may be graded without being re-run.
- **Reproducibility:** Use the specified random seeds where provided. Your results should be reproducible when the notebook is executed in the intended environment.
- **Deadline:** The submission deadline and late submission policy will be announced on SUCourse.

Grading

Component	Points
Completed: implementation, results, visualizations, and analysis	20
Homework-based quiz	30
Live demo: explanation of code and results	30
Demo questions: conceptual and applied questions	20
Total	**100**

The notebook and homework-based quiz account for **50%** of the grade. The live demo and demo questions account for the remaining **50%**. Quiz and demo arrangements will be announced separately.

1 Dataset and Data Splitting

CIFAR-10 is an image classification dataset containing 60,000 color images: 50,000 training images and 10,000 test images. Each image has a resolution of 32×32 pixels with three RGB channels. Pixel values range from 0 to 255.

Each image belongs to one of ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The classification task is to predict an image's class from its pixel values.

To keep computation manageable, use 10,000 images in total: 6,000 for training, 2,000 for validation, and 2,000 for testing.

Perform all subset selection and data splitting inside the notebook. Use train_test_split with random_state=42, shuffling, and stratification for each selection and split. Do not use subsets or splits prepared outside the notebook.

Complete the following tasks in the notebook:

1. Load CIFAR-10.
2. Select 8,000 images from the official training set, then split them into 6,000 training and 2,000 validation images.
3. Select 2,000 images from the official test set.
4. Use shuffling, stratification based on class labels, and `random_state=42` for all selections and splits.
5. Keep the test set separate until final model evaluation.
6. Print the shapes and class distributions of the resulting datasets.

2 Exploratory Data Analysis

1. Plot the class counts for the training set.
2. Calculate the global mean and standard deviation of training pixel values, as well as the mean pixel intensity for each class.
3. Display at least one example image from each class with its corresponding label.
4. Briefly describe what your plots and statistics reveal about the dataset.

3 Preprocessing

1. Normalize pixel values to $[0, 1]$ by converting them to `float32` and dividing by 255.
2. Flatten the images into feature vectors suitable for scikit-learn.
3. Report the training-set pixel mean and standard deviation before and after normalization.
4. Explain whether this normalization changes nearest-neighbor rankings under Euclidean or Manhattan distance.

4 k-NN Model Development

4.1 Hyperparameter Selection

1. Evaluate $k \in \{1, 3, 5, 7\}$ using both Euclidean distance.
2. Record the validation accuracy for all four values of k in a table.
3. Plot validation accuracy against k, in the same figure.
4. Select the best combination based on validation accuracy. Explain your selection. If multiple combinations achieve the same accuracy, state your tie-breaking rule.
5. Briefly discuss how changing k affects the behavior of the classifier. What differences would you generally expect between small and large values of k?

4.2 Final Evaluation

1. Combine the preprocessed training and validation splits into a final training set of 8,000 images.
2. Train the final k-NN classifier using the selected hyperparameters and the combined training data.
3. Evaluate the final classifier **once** on the test set and report the test accuracy.
4. Visualize the test confusion matrix. Discuss which classes appear easiest and hardest to distinguish.
5. Measure training (`fit`) and prediction (`predict`) times. Explain the difference in the context of k-NN.

5 Error Analysis

1. Identify the three most frequently confused **unordered** class pairs. For each pair, add the two corresponding off-diagonal confusion matrix counts.
2. For each selected class pair, display up to five randomly selected misclassified test images involving that pair. Include true and predicted labels. Use a fixed random seed when sampling.

   If fewer than five examples are available, display all available examples.
3. Discuss possible reasons for these mistakes based on image appearance and the behavior of distance-based classifiers.
4. Consider the representation used by the classifier. Why might two images that look similar to a human receive different predictions from k-NN?

Demo Guidelines

Each student will attend an individual demo. Demo questions will assess your understanding of the homework and whether you completed the work yourself. Questions may therefore refer to the assignment topic, code, or homework specification beyond the specific code in your notebook.

- Bring your completed notebook with all code cells and outputs already executed.
- Be prepared to navigate your notebook and explain the main implementation steps.
- You may be asked to explain a selected code cell, interpret an experimental result, or describe the effect of a small change to your implementation.
- You should understand the purpose of the training, validation, and test sets, stratification, scaling, and data leakage.
- You should be able to explain the selected value of k, the distance metric, evaluation scores, confusion matrix, and computational costs.

Questions and Discussion:

For general questions or clarifications regarding the assignment, please use the SUCourse Discussion Board so that all students can benefit from the answers. Do not share code, solutions, or completed parts of the assignment in public discussions.

Academic Integrity and AI Use

This is an individual assignment. You may discuss general concepts with classmates, but your code, experiments, figures, and written explanations must be your own work.

The following rules apply:

- Do not copy or share code, completed notebooks, experimental results, or written answers with other students.
- Do not use solutions from previous offerings of the course or publicly available completed assignments.
- AI tools may be used to clarify concepts or assist with debugging. Submitting an AI-generated solution without independently developing and understanding the implementation is not permitted. See the *Demo Guidelines* section.
- You must be able to explain every important part of your submitted notebook, including the code, preprocessing decisions, experimental results, and conclusions.

> **Important**
> Students are responsible for understanding all code and results in their submitted notebooks. The demo will assess individual understanding. Demo scheduling details will be shared on SUCourse.